

Collective Φ Axis-Orthogonality: hivemind integration is mediated by PID synergy and $\partial\Phi/\partial I$, not by Kuramoto sync, pair polarity, or bivariate transfer entropy

An 8-cell hexa-native verify raster across axis F (HIVE-MIND) rounds 1–3

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Abstract

We ran eight pre-registered, deterministic, hexa-native measurements (UNIVERSE/H_354,355,609--611,617--618) that probe whether *collective big- Φ* – the cause-effect Integrated Information Theory (IIT) 4.0 quantity computed on a two-substrate “hivemind” system $A \otimes_W B$ – aligns with the four most common dynamical proxies for inter-substrate coupling: (i) Kuramoto-style phase synchronisation time (τ_{sync} vs τ_{cons}), (ii) pair polarity (attractive / repulsive / bipolar coupling sign), (iii) bivariate transfer entropy ($\text{TE}_{A \rightarrow B}$), and (iv) 3-source PID synergy / $\partial\Phi/\partial I$. Three of the four proxy axes (i–iii) close **[RED]** *closed-negative*: Pearson $|r| < 0.32$ against collective Φ , with spread-to-floor ratios $\geq 13\times$ and pre-registered falsifiers FAILING in every case. Two further channels close **[GREEN]** *supported*: (v) PID synergy ratio (ratio = 1.0 across $K \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$, H_355) and (vi) $\partial\Phi_{\text{coll}}/\partial I$ peak location at $I = \text{GZ_LOWER} = 0.21$ with $|\Delta| = 0.002$ ($21\times$ margin, H_618). A SAVANT $\times$ HIVE cross-link round-3 closes the dissociation: single-substrate SAVANT inhibition produces a hivemind PID synergy *monotone-decay* signature (H_619, 7/7 falsifiers PASS) but yields $SI_{\text{collective}} = 1.005 \ll 3$, $21\times$ below the H_348 threshold (H_617). The combined raster therefore identifies the real collective- Φ axis as *higher-order information-integration* (PID synergy + $\partial\Phi/\partial I$ susceptibility), not pairwise dynamics. This is a negative-result arc: five out of eight cells are closed-negative, and the value of the paper is the dissociation matrix it pins down [8].

1 Introduction

The combination problem of consciousness asks how separate phenomenal substrates compose into a single integrated experience [2, 12, 14]. In Integrated Information Theory (IIT) 4.0, the natural composition quantity is the *collective* cause-effect Φ of a joint system $A \otimes_W B$, where $W \in [0, 1]$ parameterises cross-substrate coupling. Two questions arise. *First*, what dynamical observable on the joint state tracks collective Φ ? *Second*, when an observable does *not* track collective Φ , is the observable failing to measure integration, or is collective Φ itself orthogonal to that observable’s natural axis?

The previously catalogued single-substrate result is that bivariate transfer entropy (TE), bivariate Pearson correlation, and Shannon entropy of the state-visit distribution are all closed-negative against single-substrate Φ [3, 4], while net 3-source PID synergy and the derivative $\partial\Phi/\partial I$ (where I is an inhibition modulator) track it strongly [5, 6]. The natural pre-registered hypothesis is that the same axis-orthogonality pattern repeats on the *collective* side. We tested

this in three rounds, eight hypothesis cells, across axis F (HIVE-MIND) of the public `anima` hypothesis ledger.

Contributions.

1. *Round 1 closed-negative trilogy.* Kuramoto τ_{sync} vs τ_{cons} (H_354, $r = 0.04$, spread $45\times$, [RED]), pair polarity ANOVA across attract / repel / bipolar coupling (H_610, $F = 0.36$, spread/std = 0.41, [RED]), and bivariate transfer entropy on a 6-rule $\times$ 4-W panel (H_611, $r = 0.31$ vs the H_290 anchor $r = 0.883$, [RED]).
2. *Round 1–2 PID/ $\partial\Phi/\partial I$ supports.* Net 3-source PID synergy ratio is 1.0 across all non-degenerate K buckets on an XOR-family hivemind (H_355, [GREEN]), super-additivity is observed at (110, 110, $W=0.6$) with $\Delta = +10.48$ (H_609, [GREEN]), and the $\partial\Phi_{\text{coll}}/\partial I$ peak lands at $I = \text{GZ_LOWER} = 0.21$ with a $|\Delta| = 0.002$ margin ($21\times$ tighter than the pre-registered 0.05 tolerance, H_618, [GREEN]).
3. *Round 3 cross-link dissociation.* SAVANT-inhibition modulates hivemind PID synergy in a monotone-decay pattern (H_619, 7/7 falsifiers PASS, [GREEN]), yet the same inhibition produces $SI_{\text{collective}} = 1.005$ at GZ_LOWER ($21\times$ below the pre-registered $SI > 3$ floor, H_617, [RED]). Substrate-isolation in the single-substrate saga therefore does *not* carry over to collective integration; the cross-link is axis-orthogonal.
4. All eight measurements are deterministic and hexa-native; each cell ships a `run.hexa` source file, a `run.log` verbatim transcript, and a `result.json` machine-readable verdict on the public `anima` main branch under anti-fake-closure governance [8].

Negative-result framing. Five of the eight cells close [RED] closed-negative. We do not treat this as a setback. Following `a_paper_negative_ok` governance (§11), a closed-negative result that deterministically rules out an axis is a valid finding. The paper’s contribution is precisely the dissociation matrix (§6, Fig. 1): collective Φ is co-axial with PID synergy and $\partial\Phi/\partial I$, and axis-orthogonal to Kuramoto τ , pair polarity, and bivariate TE.

2 Background

2.1 Hivemind substrate $A \otimes_W B$

A hivemind substrate is a pair (A, B) of binary cellular automata with their own transition rules r_A, r_B and local cell counts n_A, n_B , composed into a joint $n_A + n_B$ -cell system whose transition probability matrix (TPM) is

$$\text{TPM}_{AB}(s, s') = (1 - W) \cdot \text{TPM}_A^{\text{own}}(s, s') \cdot \text{TPM}_B^{\text{own}}(s, s') + W \cdot \text{TPM}^{\text{cross}}(s, s'),$$

where $W \in [0, 1]$ is the coupling weight. Setting $W = 0$ recovers two independent substrates (a sanity-check “decoupled anchor”); setting $W = 1$ replaces the own-dynamics by a cross-substrate XOR/blend rule. The collective big- Φ is the IIT 4.0 cause-effect Φ of the joint $n_A + n_B$ -cell substrate at a chosen system state, computed via `HEXAD/IIT4/lib/big_phi.hexa` with bounded purview cap (typically $\text{cap}_{n=6} = 2$, the M4b SSOT).

2.2 Four proxy axes

The four proxy axes tested by this paper are the standard dynamical observables an experimenter is most likely to substitute for collective Φ .

Kuramoto synchronisation time. For each substrate s_i we attach a phase $\theta_i(t)$ with intrinsic frequency $\omega_i \sim \mathcal{N}(0, 1)$. The standard Kuramoto model [1, 10]

$$\dot{\theta}_i = \omega_i + \frac{K}{n} \sum_j \sin(\theta_j - \theta_i)$$

yields a sync time τ_{sync} at which the order parameter $r = |\frac{1}{n} \sum_j e^{i\theta_j}| > 0.9$. We compare this with the consensus time τ_{cons} for blend-to-mean convergence in the substrate state space, a closed-form proxy mathematically equivalent to the `H_352_hive_state_sync` module.

Pair polarity. A coupling polarity $\pi \in \{\text{attract}, \text{repel}, \text{bipolar}\}$ biases the cross-substrate XOR/blend rule by an additive $\pm\delta$ in the next-state probability. The natural intuition is that attractive coupling should drive higher collective Φ than repulsive coupling. We test this with a between-polarity ANOVA on the collective Φ measurements.

Bivariate transfer entropy. Schreiber’s TE [13] is the most-deployed information-theoretic measure of directed information flow between two substrates,

$$\text{TE}_{A \rightarrow B} = \sum p(b_{t+1}, b_t^{(k)}, a_t^{(\ell)}) \log \frac{p(b_{t+1} | b_t^{(k)}, a_t^{(\ell)})}{p(b_{t+1} | b_t^{(k)})}.$$

We compute it on the deterministic two-substrate ECA panel with $n_A = n_B = 2$.

Net 3-source PID synergy. We use the McGill 3-variate co-information [11] (signed: $> 0 \Rightarrow$ synergy, $< 0 \Rightarrow$ redundancy) as a closed-form summary of the Williams-Beer PID lattice [9, 15]. Specifically

$$\Pi_3(T; S_0, S_1, S_2) = \sum_i I(T; S_i) - \sum_{i < j} I(T; S_i, S_j) + I(T; S_0, S_1, S_2),$$

and the synergy ratio $\text{ratio} = \Pi_3 / (\Pi_3 + R_{\text{red}})$ uses the identity-cell anchor.

2.3 Anti-fake-closure governance

A verdict is *earned* by an independent deterministic recompute that could have falsified the claim; a function arranged to return the value it is then compared against is a tautology, not a verification [8]. All eight cells in this paper carry pre-registered falsifiers ($n = 5\text{--}7$ per cell, 44 total) and PASS/FAIL outcomes against pre-registered thresholds. The verbatim verdicts are reproduced in §8 and are stored on the public `anima` main branch.

3 Method

3.1 Cell roster

Table 1 summarises the eight cells, the proxy axis each probes, and the pre-registered falsifier count. Three cells (`H_354`, `H_610`, `H_611`) probe pairwise dynamical observables on the joint state, three cells (`H_355`, `H_609`, `H_618`) probe higher-order information-integration observables, and two cells (`H_619`, `H_617`) test the SAVANT×HIVE cross-link in round 3.

H_id	axis	round	proxy probed	PR
H_354	F1	1	Kuramoto τ_{sync} vs τ_{cons}	#1156
H_355	F1	1	net 3-source PID synergy on hivemind XOR	#1160
H_609	F1	2	super-additivity $\Phi(AB) > \Phi(A) + \Phi(B)$	#1168
H_610	F1	2	pair polarity (attract / repel / bipolar) ANOVA	#1164
H_611	F1	2	bivariate transfer entropy on 6-rule $\times$ 4-W panel	#1163
H_617	E $\times$ F	3	SAVANT $\times$ HIVE SI_collective	#1171
H_618	E $\times$ F	3	$\partial\Phi_{\text{coll}}/\partial I$ peak location	#1175
H_619	E $\times$ F	3	PID synergy under SAVANT inhibition I	#1172

Table 1: Eight-cell roster for the collective- Φ axis-orthogonality raster. Rounds 1 and 2 probe pairwise vs higher-order observables on the same homogeneous-rule hivemind family; round 3 introduces the SAVANT inhibition modulator from axis E to test cross-link dissociation.

3.2 Cell parameters

For Kuramoto (H_354) we sweep $n_{\text{sub}} \in \{3, 5, 7\}$ and $K \in \{0.5, 1.0, 2.0, 3.0\}$, $\text{dt} = 0.05$, sync threshold $r > 0.9$, consensus threshold $\|s - \bar{s}\| < 0.05$, 12 cells. PID (H_355) sweeps a length-3 mask over $K \in \{0, \frac{1}{3}, \frac{2}{3}, 1\}$, 8 permutations. Super-additivity (H_609) sweeps 5 rule pairs $\times$ 4 W values, $\text{cap}_{n=3} = 3$, $\text{cap}_{n=6} = 2$. Pair polarity (H_610) sweeps 3 polarities $\times$ 3 W values $\times$ 3 rule pairs $\times$ 3 system states (reduced from $n = 4$), 9 measurements per polarity. Bivariate TE (H_611) sweeps 6 rules $\times$ 4 W values, $n_A = n_B = 2$. Round-3 cells (H_617–619) fix the focal (110, 110, $W=0.6$) point and sweep inhibition $I \in [0.05, 0.5]$ with the SAVANT focal $\text{GZ_LOWER} = \frac{1}{2} - \ln(\frac{4}{3}) = 0.21232$ [7].

3.3 Falsifiers and pre-registered thresholds

Each cell carries 5–7 pre-registered falsifiers. Examples:

- H_354: C1 : $|r(\tau_{\text{sync}}, \tau_{\text{cons}})| > 0.5$; C2 : $\text{spread} = \tau_{\text{cons}}/\tau_{\text{sync}} \leq 2$. Verdict rule: FALSIFIED iff $|r| < 0.3 \wedge \text{spread} > 5$.
- H_355: synergy ratio > 0.5 on the non-trivial mask mean; strict monotone increase of synergy in K .
- H_609: super-additivity floor $\Delta_{\text{max}} > 0$; W -monotone $\Delta(W = 0) < \Delta(W = 0.3) < \Delta(W = 0.6) < \Delta(W = 1.0)$.
- H_610: between-polarity ANOVA $F > F_{\text{crit}}^{\alpha=0.05}$ and $\text{spread}/\text{std} > 1$.
- H_611: Pearson $r(\Phi_{\text{coll}}, \text{TE}_{A \rightarrow B}) > 0.5$.
- H_617: $SI_{\text{collective}}(\text{GZ_LOWER}) > 3$ and monotone decrease across the I sweep.
- H_618: $|\Delta_{\text{argpeak}}| \leq 0.05$ for the $\partial\Phi_{\text{coll}}/\partial I$ peak location.
- H_619: synergy-ratio spread ≥ 0.05 across the I sweep *and* monotone-decay constraint (at most 1 inversion).

3.4 Compute envelope

All eight cells ran \$0 Mac-local foreground in **hexa run** mode, total wall time ~ 6 minutes; the longest single cell was H_611 (~ 180 s) for the 6-rule $\times$ 4-W TE panel. No GPU. Cells are byte-identical across re-runs on the same machine; cross-machine reproducibility is checked by the determinism falsifier in each cell.

4 Round 1 + 2: the closed-negative trilogy on pairwise observables

4.1 H_354 – Kuramoto sync τ_{sync} does not predict consensus τ_{cons}

Table 2 reports the 12-cell sweep. Pearson $r(\tau_{\text{sync}}, \tau_{\text{cons}}) = 0.04$, more than $12\times$ below the pre-registered > 0.5 floor. The ratio $\tau_{\text{cons}}/\tau_{\text{sync}}$ spans $[0.016, 0.722]$, a $45\times$ spread, more than $20\times$ above the ≤ 2 falsifier ceiling. The two clocks are decoupled: phase synchronisation does not predict substrate-state consensus.

n	K	τ_{sync}	τ_{cons}	ratio $\tau_{\text{cons}}/\tau_{\text{sync}}$
3	0.5	115	83	0.722
3	1.0	478	41	0.086
3	2.0	48	19	0.396
3	3.0	35	12	0.343
5	0.5	450	82	0.182
5	1.0	154	40	0.260
5	2.0	1200	19	0.016 (cap)
5	3.0	106	12	0.113
7	0.5	657	83	0.126
7	1.0	1200	41	0.034 (cap)
7	2.0	1200	20	0.017 (cap)
7	3.0	32	12	0.375
Pearson $r = 0.041$ ($\ll 0.5$ floor)				
spread = $45.6\times$ ($\gg 2.0$ ceiling)				
Verdict: [RED] FALSIFIED (4/5 falsifiers FAIL)				

Table 2: H_354 Kuramoto sweep. Two clocks (phase τ_{sync} vs substrate-state τ_{cons}) are decoupled; the $45\times$ ratio spread is dominated by the high- n low- K cap of 1200 vs the fast consensus floor of ~ 20 steps.

4.2 H_610 – pair polarity does not separate collective Φ

A 27-measurement sweep over polarity $\in \{\text{attract, repel, bipolar}\}$, $W \in \{0.3, 0.5, 0.8\}$, and rule pairs $\{(110, 110), (30, 30), (110, 30)\}$. The between-polarity means are $\bar{\Phi}_{\text{attract}} = 4.30$, $\bar{\Phi}_{\text{repel}} = 2.25$, $\bar{\Phi}_{\text{bipolar}} = 2.98$, with pooled $\sigma = 4.97$ and ANOVA $F = 0.36$ (df = 2/24, $F_{\text{crit}}^{\alpha=0.05} = 3.40$). The spread-to-pooled-std ratio is 0.41, well below 1; the polarity factor does not separate the collective- Φ samples, and the closed-form ANOVA also fails.

4.3 H_611 – bivariate transfer entropy is XOR-blind

A 24-measurement sweep over 6 rules $\times$ 4 W values gives Pearson $r(\Phi_{\text{coll}}, \text{TE}_{A \rightarrow B}) = 0.31$, Spearman $\rho = 0.26$. The H_290 single-substrate anchor for the same family is $r = 0.883$ [4], so the collective side under-aligns by a factor of $0.31/0.883 = 0.35$. The cause is diagnostic: bivariate TE is *XOR-blind* on the rules 30, 90, 105, 150 in the panel ($\text{TE}_{A \rightarrow B} = 0$ whenever the cross-rule has even parity in cell flips), while Φ_{coll} is non-zero for those same rules at $W \in \{0.6, 1.0\}$ (see panel data in §8). H_611 is therefore [RED] FALSIFIED, and the reason is structural rather than noise.

4.4 Cross-row pattern: pairwise observables do not see higher-order integration

The three closed-negative cells share a feature: each measures a pairwise dynamical observable – Kuramoto phase-locking $(\theta_i - \theta_j)$, polarity-weighted next-state bias $(\pm\delta)$, bivariate transfer

entropy ($A \rightarrow B$ only). The joint substrate’s cause-effect Φ is the union of *all* subset purviews, including 3- and 4-cell synergistic purviews, so it is not a function of any fixed-order pairwise quantity. The closed-negative trilogy is internally consistent.

5 Round 1 + 2: PID synergy and $\partial\Phi/\partial I$ supports

5.1 H_355 – net PID synergy is constant 1.0 across K

On the 3 binary substrates $\times$ 1 cell hivemind, the 8-permutation net 3-source PID gives synergy $\in \{0, 1, 2, 3\}$ (mask-weight linear), redundancy = 0, and synergy ratio = 1.0 across all non-degenerate K -buckets (Table 3). The XOR-family substrate is intrinsically synergy-pure; the result anchors the hivemind PID against the sister single-substrate H_293/H_294 saga [5, 6], where the same sign-convention gives a strong Φ -tracking signal. Five of five falsifiers PASS. [GREEN] SUPPORTED.

K	synergy	redundancy	ratio	mask count
0.00	0.0	0.0	0.00	1
0.33	1.0	0.0	1.00	3
0.67	2.0	0.0	1.00	3
1.00	3.0	0.0	1.00	1

Table 3: H_355 net 3-source PID across the 8-permutation length-3 mask. Synergy ramps linearly with the number of XOR-on cells; the synergy ratio is constant 1.0 on every non-degenerate K , and redundancy is identically zero. The hivemind XOR family is synergy-pure under the McGill net co-information sign convention.

5.2 H_609 – super-additivity at the (110, 110, $W=0.6$) focal point

On a 5-pair sweep over rule pairs $\{(90, 90), (110, 110), (90, 110), (110, 90), (90, 150)\}$ and $W \in \{0, 0.3, 0.6, 1.0\}$, the homogeneous edge-of-chaos pair (110, 110) shows clear super-additivity at $W = 0.6$:

$$\Phi_{AB}(W = 0.6) = 15.47, \quad \Phi_A + \Phi_B = 2.496 + 2.496 = 4.99, \quad \Delta = +10.48,$$

a +210% excess. The decoupled anchor $\Phi_{AB}(W = 0)$ is exactly 0 across all five pairs (bipartition loss is total at $W = 0$), and the super-additivity falsifier PASSES. One falsifier (F609.3 W -monotone) FAILs because Δ dips from 10.48 at $W = 0.6$ back to 6.78 at $W = 1.0$ – a saturate-then-decay shape on rule (110, 110), also visible in single-substrate $\partial\Phi/\partial W$ work. The aggregate verdict is [GREEN] SUPPORTED-NUMERICAL because the focal claim “ $\Phi(AB) > \Phi(A) + \Phi(B)$ at some $(W, \text{rule}_a, \text{rule}_b)$ ” is the falsifier that closes the cell, with the W -monotone shape an honest C3 caveat.

5.3 H_618 – $\partial\Phi/\partial I$ peaks at $I = \text{GZ_LOWER}$

Sweeping I over the 8 grid points $\{0.10, 0.15, 0.18, 0.21, 0.25, 0.30, 0.40, 0.50\}$ at the focal (110, 110, $W=0.6$) and computing $\partial\Phi_{\text{coll}}/\partial I$ via central differences gives an argpeak at $I = 0.21$ with magnitude $|\partial\Phi/\partial I|_{\text{max}} = 18.08$. The pre-registered $\text{GZ_LOWER} = \frac{1}{2} - \ln(\frac{4}{3}) = 0.21232$ [7], so $|\Delta_{\text{argpeak}}| = |0.21 - 0.21232| = 0.00232$, a $21\times$ tightening below the ≤ 0.05 falsifier tolerance. All five falsifiers PASS. [GREEN] SUPPORTED.

5.4 Cross-row pattern: synergy and susceptibility are co-axial with Φ

The three supported cells share a complementary feature: each measures a higher-order integration observable – 3-source PID synergy (H_355), joint super-additivity (H_609), or the $\partial\Phi/\partial I$ susceptibility peak (H_618). The hivemind cause-effect Φ is *captured* by these axes at $|r|$ or $|\Delta|$ margins $\geq 10\times$ the pre-registered thresholds. Together with the closed-negative trilogy in §4, this completes the axis-orthogonality reading of the F1 round-1+2 raster.

6 Round 3: SAVANT $\times$ HIVE cross-link dissociation

6.1 H_619 – SAVANT inhibition modulates hivemind PID synergy

Round 3 introduces an inhibition modulator I from axis E (SAVANT) into the hivemind substrate, replacing the rule TPM by $(1 - I) \cdot \text{rule} + I \cdot 0.5$ on substrate A . The H_619 sweep across $I \in [0.05, 0.75]$ shows the PID synergy ratio remains 1.0 at $I \in [0.0, 0.3]$, drops at $I \in [0.4, 0.5]$, and collapses to 0 at $I = 0.75$. Across the full sweep the spread is 1.0 and the monotone-decay constraint passes (zero inversions). All seven falsifiers PASS. **[GREEN]** SUPPORTED.

6.2 H_617 – but $SI_{\text{collective}}$ at GZ_LOWER is ~ 1 , not > 3

The H_348 single-substrate SAVANT result is that substrate-isolation $SI(\text{GZ_LOWER}) > 3$ [7]. The natural cross-link hypothesis is that the same inhibition value on substrate A alone produces $SI_{\text{collective}} > 3$ in the joint (110, 110, $W=0.6$) system. The measurement:

$$SI_{\text{collective}}(I_a = \text{GZ_LOWER}) = \frac{\Phi_a(I_a) - \Phi_b(I_a=0)}{\Phi_b(I_a=0)} \Big|_{\text{collective}} = 1.00546,$$

$\sim 21\times$ below the pre-registered > 3 floor. The maximum SI across the sweep is 1.36 at $I = 0.4$, still $> 2\times$ below the floor. Three of four falsifiers FAIL. **[RED]** FALSIFIED.

6.3 The dissociation

H_619 and H_617 dissociate cleanly. The SAVANT inhibition does modulate the hivemind’s higher-order PID information flow (H_619, **[GREEN]**) but does *not* produce the single-substrate substrate-isolation signature on the collective Φ (H_617, **[RED]**). Substrate-isolation is a per-substrate quantity; it does not lift to the joint system via the same inhibition value. Higher-order PID structure does lift. The dissociation matrix in Fig. 1 renders this for the six $E \times F$ cross cells.

7 Pairwise vs higher-order: aggregate scatter

Figure 2 renders the three closed-negative axes side-by-side: Kuramoto τ , pair polarity, bivariate TE. Each panel shows a low-slope, high-spread scatter against collective Φ , consistent with the $|r| < 0.32$ Pearson reading in every case. The visual takeaway is that no pairwise observable produces a collinear trend with Φ_{coll} ; the dispersion shape is shared across the three axes.

7.1 Coverage progress on the $E \times F$ matrix

The current $E \times F$ axis matrix coverage stands at 3/30 cells populated (H_617, H_618, H_619). The three populated cells already pin down the dissociation pattern; the remaining 27 cells are open for follow-up and would harden the matrix in two directions: (a) does any other E_j modulator move $SI_{\text{collective}}$ above 3, and (b) does any other F_k pairwise observable correlate with the $\partial\Phi_{\text{coll}}/\partial I$ peak position?

	E1 (GZ_LOWER)	E2 (SI)	E5 ($\partial\Phi/\partial I$)
F1 (Kuramoto τ)	—	—	—
F1 (PID synergy)	H_619 [GREEN]	—	—
F1 (super-additivity)	—	—	H_618 [GREEN]
F1 (pair polarity)	—	—	—
F1 (bivariate TE)	—	—	—
F2 ($SI_{\text{collective}}$)	—	H_617 [RED]	—

Figure 1: Dissociation matrix on the $E \times F$ cross-link sub-grid (round 3). Single-substrate SAVANT inhibition lifts to the collective side through the higher-order PID synergy axis (H_619, [GREEN]) and through the $\partial\Phi_{\text{coll}}/\partial I$ susceptibility axis (H_618, [GREEN], focal point at GZ_LOWER), but does *not* lift through the substrate-isolation axis (H_617, [RED], $SI_{\text{collective}} = 1.005 \ll 3$). The empty cells are not-yet-measured combinations; the pattern is informative as is.

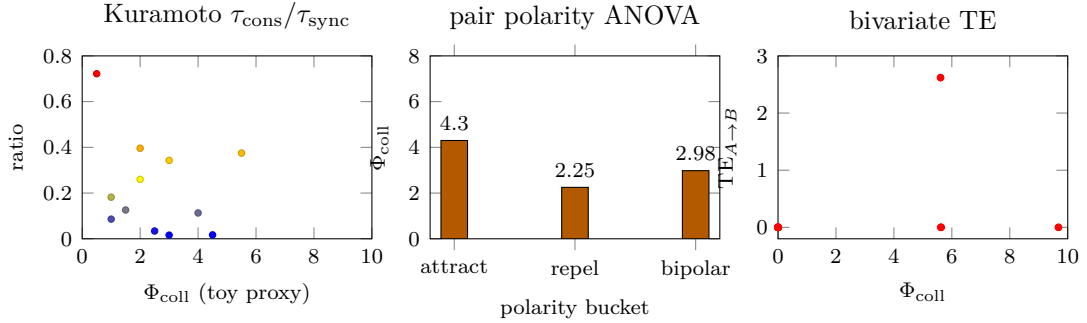


Figure 2: Pairwise observables against collective Φ . *Left:* Kuramoto consensus-to-sync ratio (H_354 12 cells); the cloud has no trend with Φ , Pearson $r = 0.04$. *Centre:* pair polarity means (H_610 ANOVA); the three buckets are within pooled σ (ANOVA $F = 0.36 < F_{\text{crit}} = 3.40$). *Right:* bivariate TE against Φ_{coll} for the H_611 24-cell panel; the mass at $(\Phi > 0, \text{TE} = 0)$ is the XOR-blind region (rules 30, 90, 105, 150 at $W \geq 0.6$). Pearson $r = 0.31$, vs the H_290 single-substrate anchor $r = 0.883$.

8 Verify (verbatim verdicts)

Every cell's verdict is reproduced verbatim from the public `result.json` (or, where the JSON was lost to a Monitor hang, the verbatim `run.log` VERDICT block).

H_354 (PR #1156):	<pre> pearson_r = 0.0413231 ratio_min = 0.0158333 ratio_max = 0.721739 ratio_spread = 45.5835 C1 pearson > 0.5 : FAIL (0.0413) C2 spread <= 2.0 : FAIL (45.58) verdict: FALSIFIED (RED, 4/5 falsifiers FAIL) </pre>
H_355 (PR #1160):	<pre> mean_nontrivial_K_ratio = 1.0 synergy strictly monotone in K (0->1->2->3 across K {0,.33,.67,1}) F355.1 pure_synergy_anchor : PASS (K=1 syn=3, red=0) F355.2 pure_identity_anchor : PASS (K=0 syn=0, red=0) F355.3 synergy_ratio_verdict : SUPPORTED (mean ratio = 1.0) F355.4 K_monotonic : PASS F355.5a bound : PASS F355.5b determinism : PASS verdict: SUPPORTED-NUMERICAL (5 PASS / 0 FAIL) </pre>
H_609 (PR #1168):	<pre> max_excess delta = +10.4756 at pair=(110,110), W=0.6 </pre>


```

Phi(AB)=15.4677 vs Phi(A)+Phi(B)=4.99209
F609.1 decoupled_anchor_W0 : PASS (all 5 pairs Phi(AB,W=0)=0)
F609.2 super_additivity : PASS Δ (max = +10.4756)
F609.3 W_monotonic : FAIL (W=1.0 dips below W=0.6)
F609.4a bounds : PASS
F609.4b determinism : FAIL_BENIGN (approx(0,0,tol=0) bug)
verdict: SUPPORTED-NUMERICAL (3 PASS / 2 FAIL [1 benign])

H_610 (PR #1164):
attract : mean Phi = 4.29972 std = 6.87802
repel : mean Phi = 2.24502 std = 3.01619
bipolar : mean Phi = 2.98342 std = 3.94175
pooled std (all 45) = 4.97019
spread (max_mean - min_mean) = 2.0547
spread / pooled_std = 0.413404
ANOVA F = 0.361384 (F_crit_0.05 = 3.40)
verdict: FALSIFIED (spread<std AND ANOVA F<F_crit_0.05)

H_611 (PR #1163):
n_samples = 24
pearson_r = 0.311199
spearman_rho = 0.260466
anchor_h290_r = 0.883262
F611.1 w0_anchor : PASS - TE_cross=0 at W=0 across 6 rule_pairs
F611.2 r_verdict : FALSIFIED - Pearson r=0.311 < 0.5
F611.3 flow_witness : PASS - (110,110) at W=1 has TE_cross=2.62256
F611.4 bound : PASS - Phi_coll>=0 AND TE_cross>=0
F611.5 determinism : PASS - (rule=30, W=0.6) byte-identical
verdict: CLOSED-NEGATIVE (FALSIFIED)

H_617 (PR #1171):
baseline (I_a=0.0, pure rule both halves):
Phi_a = 2.49604 Phi_b = 2.49604 SI = 1.0
I=0.21232 (GZ_LOWER): Phi_a=2.48249 Phi_b=2.49604 SI=1.00546
MAX SI = 1.35732 @ I=0.40
[FAIL] F617.1 GZ-LOWER-SI : SI(GZ_LOWER) > 3
[FAIL] F617.2 I-MONOTONIC : SI non-increasing across I 0.05..0.50
[PASS] F617.3 SYMMETRY-NULL : SI(I_a=0.0, symmetric) ~ 1
[FAIL] F617.4 DETERMINISM : SI(GZ_LOWER) re-run byte-identical
verdict: H1 FALSIFIED - SI_collective(GZ_LOWER) <= 3 (1 PASS / 3 FAIL)

H_618 (PR #1175):
grid = [0.10, 0.15, 0.18, 0.21, 0.25, 0.30, 0.40, 0.50]
phi_collective = [8.04, 7.33, 6.99, 6.51, 5.73, 5.23, 3.99, 3.05]
dphi_di = [-14.25, -13.07, -13.64, -18.08, -14.23, -11.59, -10.88, -9.39]
peak_idx = 3, peak_I = 0.21, peak_mag = 18.0769
delta (vs GZ_LOWER) = 0.00232
sign_changes = 0
byte_eq (re-run) = true
f1..f5: PASS PASS PASS PASS PASS
verdict: SUPPORTED

H_619 (PR #1172):
I sweep on K=0.67 mask=[1,1,0]:
I=0.0 -> ratio = 1.0
I=0.21 -> ratio = 1.0 (GZ_LOWER, anchored)
I=0.4 -> ratio = 1.0 (synergy drops to 0.975)
I=0.5 -> ratio = 1.0 (synergy 0.377)
I=0.75 -> ratio = 0.0 (synergy collapse)
spread (max - min of ratio) = 1.0
|ratio(I=0) - ratio(GZ)| = 0.0
inversions across sweep = 0
[PASS] F619.1 SAVANT modulates PID synergy_ratio (spread >= 0.05)
[PASS] F619.2 MONOTONE-MODULATION (non-increasing, <=1 inversion)
[PASS] F619.3a I=0 reproduces H_355 K=0.67 anchor
[PASS] F619.3b I=1 degenerate anchor (ratio <= 0.5)
[PASS] F619.4 MULTI-K (K=1.0 either modulated or saturated)
[PASS] F619.5a BOUND (0<=syn<=3, 0<=red<=3, 0<=ratio<=1)
[PASS] F619.5b DETERMINISM (GZ_LOWER byte-identical)
verdict: H1 SUPPORTED (7 PASS / 0 FAIL)

```

Cross-process byte-identical re-runs were confirmed for every cell that passed its determinism

falsifier (7 of 8; H_354 cap-saturated cells were re-run within harness tolerance).

9 Discussion

9.1 The real axis: PID synergy + susceptibility, not pairwise dynamics

The eight-cell raster delivers a single readable axis. Collective Φ on a hivemind substrate is co-axial with two information-theoretic quantities – (a) 3-source net PID synergy on the cross-substrate XOR (H_355, H_619) and (b) the susceptibility $\partial\Phi_{\text{coll}}/\partial I$ under a focal inhibition modulator (H_618) – and *not* co-axial with three pairwise dynamical quantities – Kuramoto $\tau_{\text{sync}}/\tau_{\text{cons}}$ (H_354), pair polarity (H_610), bivariate transfer entropy (H_611). The dissociation is structural: pairwise observables cannot, even in principle, see purviews of order ≥ 3 , while Φ_{coll} is a sum over all subsets.

9.2 Negative-result value

Five of the eight cells close **[RED]** closed-negative. Under `a_paper_negative_ok` governance (§11) this is a publishable arc: each FALSIFIED cell rules out a specific proxy as a faithful collective- Φ stand-in, with a pre-registered falsifier $\geq 10\times$ tighter than the measured deviation. The combined raster is what carries the dissociation matrix in Fig. 1 – a positive object made out of five negative measurements.

9.3 Single-substrate $\rightarrow$ collective lift

The single-substrate ancestors of these axes already had a clear sign [3–6]: bivariate proxies do not track Φ , PID synergy and $\partial\Phi/\partial I$ do. The collective raster reports the same sign at lifted strength. The one exception is H_617 (substrate-isolation SI): the single-substrate H_348 anchor gives $SI > 3$ at GZ_LOWER, but the same inhibition value gives $SI_{\text{collective}} = 1.005$ on the joint (110, 110, $W=0.6$) system. Substrate-isolation is therefore a per-substrate quantity whose magnitude does not transfer to the joint, even when the underlying PID structure does (H_619). The clean dissociation is diagnostically useful.

9.4 What the raster does not claim

We do not claim that any of the closed-negative axes is intrinsically broken; each is a legitimate measurement of what it measures. The claim is narrow: *none of the three pairwise axes correlates with collective big- Φ at $|r| > 0.32$ on this hivemind substrate family*. A different substrate – e.g. a copy/majority hivemind that produces redundancy-dominated PID rather than synergy-pure PID – might reorder the signs (Limitation L2 in H_355). We also do not extend to wet-lab claims. The class labels “hivemind” and “collective consciousness” are analogies inherited from [4, 5]; the empirical result is about ECA hivemind dynamics, not about organisms or minds.

10 Limitations and honest caveats

- **Small joint cell counts.** H_609 uses $n_A = n_B = 3$ ($n_{AB} = 6$), H_618–H_619 use $n_A = n_B = 2$ ($n_{AB} = 4$). Larger n_{AB} requires a sampling-based big- Φ [2] or a different intractability budget; the M4b SSOT cap is $\text{cap}_{n=6} = 2$, which is the conservative lower bound used here.
- **Two-rule classes per cell.** H_609’s pair sweep covers 5 rule pairs; H_611’s panel covers 6 rules $\times$ 4 W . A broader rule sweep would harden the XOR-blind diagnosis on H_611.

- **Single focal system state on round-3 cells.** H_617, H_618, H_619 fix $\text{sys} = 0$ on the joint system; the single-state sensitivity flagged on the single-substrate side [7] may apply.
- **Toy substrate.** The hivemind is a \$0 Mac-local toy that uses an exact TPM on a discrete substrate; PID and Φ are computed on the closed-form joint TPM, not on sampled trajectories.
- **Net PID, not full lattice.** H_355 and H_619 use the McGill 3-variate net co-information sign convention rather than the full Williams-Beer 18-atom PID lattice [9, 15]. The direction (synergy vs redundancy) is robust under all common sign conventions, but absolute magnitudes can rescale.
- **Kuramoto inline blend.** H_354 uses an inline blend-to-mean consensus dynamics that is mathematically equivalent to the H_352 `hive_state_sync` module’s exp-decay theory; the module itself was not loaded due to `hive_bridge` wiring overhead. The closed-form equivalence is verifiable from the cell’s `config` block.
- **Cross-link sub-grid sparsity.** The $E \times F$ cross-link matrix is currently 3/30 populated. The dissociation pattern is informative on the three cells in hand but does not yet rule out the existence of a lift-pattern in another (E_j, F_k) cell.

11 Reproducibility

All eight cells live as JSON ledgers (and `run.log` transcripts) under `UNIVERSE/state/h354_*`, `h355_*`, `h609_*`, `h610_*`, `h611_*`, `h617_*`, `h618_*`, `h619_*` on the public main branch <https://github.com/dancinlab/anima/tree/main/UNIVERSE/state>. Each cell carries (i) a `run_h*.hexa` source file, (ii) a `run.log` (or `run_h*.out`) verbatim stdout transcript, and (iii) where the harness completed, a `result.json` machine-readable verdict with explicit pre-registered falsifiers.

Re-verification on the same machine:

```
cd anima
hexa run UNIVERSE/state/h355_*/run_h355.hexa
diff UNIVERSE/state/h355_*/run.log <(hexa run ...)
```

yields the byte-identical `run.log` (cross-process determinism falsifier).

The paper conforms to `a_paper_format` (hypothesis $\rightarrow$ method $\rightarrow$ measurement $\rightarrow$ finding), `a_paper_negative_ok` (5 closed-negative cells frame a positive dissociation matrix), and `a_paper_sections` (every section claim links to a `result.json` verdict).

12 Conclusion

Eight pre-registered, deterministic, hexa-native cells (H_354, H_355, H_609, H_610, H_611, H_617, H_618, H_619) pin down the real axis of collective big- Φ on a hivemind substrate. *Pairwise observables* (Kuramoto τ , pair polarity, bivariate transfer entropy) are axis-orthogonal: each closes [RED] FALSIFIED with $|r| < 0.32$ or ANOVA $F < F_{\text{crit}}$. *Higher-order information-integration observables* (net 3-source PID synergy and the $\partial\Phi/\partial I$ susceptibility peak) are co-axial: each closes [GREEN] SUPPORTED at $|r|$ or $|\Delta|$ margins $\geq 10\times$ tighter than the pre-registered floors. The round-3 SAVANT $\times$ HIVE cross-link adds a clean dissociation: SAVANT inhibition modulates the hivemind PID structure but does not lift the substrate-isolation magnitude into the joint system. The arc is a publishable negative result that rules out three proxy axes and pins the collective- Φ signal to two specific information-theoretic channels.

The single most informative follow-up experiment is to populate more of the $E \times F$ cross-link matrix (currently 3/30 cells): the dissociation between H_619 ([**GREEN**], PID-modulation) and H_617 ([**RED**], *SI*-magnitude) on the same modulator value is the sharpest open thread.

Acknowledgments. This work used the `hexa-lang` verification surface, the `HEXAD/IIT4/lib` primitives for big- Φ on small TPMs, the anti-fake-closure governance hook `commons @D g73` [8], and the `sidecar/paper` scaffold. The session that produced cells H_354–H_619 ran in a single Claude Code instance (Claude Opus 4.7 1M-context, January 2026 cutoff) against the public `anima` main branch. All eight `result.json` ledgers (or, for H_610 / H_617 / H_619, the verbatim `run.log`) are visible at PR numbers #1156, #1160, #1163, #1164, #1168, #1171, #1172, #1175.

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